INCH-POUND
A-A-58004D
15 April 2009
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COMMERCIAL ITEM DESCRIPTION

TRUCK, FIRE FIGHTING, WATER TANKER, 4,000 GALLON CAPACITY

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers a commercial water tanker truck. The tanker has a 6x4 chassis with a diesel engine and an automatic transmission; a two person cab with two doors; a 4,000 gallon water tank; a 1,250 gallon per minute (gpm) fire pump; equipment compartments; and hose. The water tanker truck is intended to: store and transport water over paved roads, gravel roads, or unimproved surfaces, and draft, pump, and transfer water in support of brush, structural, and aircraft fire fighting vehicles.

2. SALIENT CHARACTERISTICS. The tanker shall be in accordance with the applicable requirements of the following chapters of National Fire Protection Association (NFPA) 1901, 2009 Edition, for Mobile Water Supply Apparatus:

Chapter	Title
1	Administration
2	Referenced Publications
3	Definitions
4	General Requirements
7	Mobile Water Supply Fire Apparatus
12	Chassis and Vehicle Components
13	Low-Voltage Electrical Systems and Warning Devices
14	Driving and Crew Areas
15	Body, Compartments, and Equipment Mounting
16	Fire Pumps and Associated Equipment
19	Water Tanks
23	Line Voltage Electrical Systems

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC 542 CSW/CBSSS, 295 Byron Street, Robins AFB, GA 31098-1611.

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2.1 Administration.

2.1.1 Environmental conditions.

2.1.1.1 Altitude. Unless otherwise specified (see 6.2), the tanker, including the pumping system, shall be designed for operation at 2,000 feet above sea level.

2.1.1.2 Temperature range. The tanker shall be capable of satisfactory storage and operation in temperatures ranging from 0° to 125° F. The tanker shall be equipped with a winterization system that extends the operational range of the tanker, the agent pump and the agent piping systems to 0° F. While in operation, the appropriate systems requiring winterization shall be powered by the tanker. While in storage, the winterization system shall be powered through the electrical shoreline connection (see 2.9.1).

2.1.2 Foreign object damage. All loose metal parts, such as pins, shall be securely attached to the vehicle with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

2.1.3 Mobility.

2.1.3.1 Operating terrain. The tanker shall operate on paved and graded gravel roads.

2.1.3.2 Acceleration. The fully loaded tanker shall accelerate from 0 to 35 miles per hour (mph) within 25 seconds on a level, paved road.

2.1.3.3 Maximum speed. The fully loaded tanker shall attain a minimum top speed of 50 mph on a level, paved road.

2.1.4 Overall dimensions. Overall dimensions shall be the minimum consistent with the operational performance and the design constraints necessary to achieve the specified performance. Overall dimensions shall not exceed:

Length	390 inches
Width	102 inches (excluding mirrors)
Height	120 inches

2.1.5 Turning diameter. The fully loaded tanker shall have a wall to wall turning diameter of less than 80 feet in both directions.

2.1.6 Angles of approach and departure. The fully loaded tanker shall have angles of approach and departure of not less than 20° as required by 5.3.2.3 of NFPA standard 1906. The rational is that the vehicle may be used for wild land fire fighting.

2.1.7 Manuals.

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2.1.7.1 Technical manuals. The overall format for the manuals may be military specification, commercial, or a combination of both. Each technical manual shall have a title page. Line art shall be used to the maximum extent possible for illustrations and parts lists.

a. The contractor shall validate the technical manuals for accuracy prior to submission to the procuring activity for verification. The contractor shall submit one complete set to the procuring activity for verification at least 60 days prior to the first production test. Any changes or corrections noted by the procuring activity shall be corrected and updated pages or manuals shall be submitted to the procuring activity.

b. Once approved by the procuring activity, the contractor shall pack two complete sets of technical manuals with each vehicle. An additional two complete sets shall be submitted to the procuring activity for stock.

c. Once approved by the procuring activity and a Technical Order number is assigned, the contractor shall pack two complete sets of both hard copy and CDs with each vehicle and one complete set of both hard copy and CDs shall be submitted to the procuring activity for stock. Mail should be delivered to the procuring agency at:

Dallas Perry
580 CBSS/GBZV
380 Richard Ray Blvd Suite 104
Robins AFB, GA 31098-1640

d. The contractor shall provide written reproduction rights to the United States Air Force for all technical manuals, to include manuals developed by the contractor as well as those for all subsystem components used in the manufacture of the vehicle. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement. This document shall be presented on official company letterhead.

2.1.7.1.1 Operator's manual. The operator's manual shall include all information required for the safe and efficient operation of the vehicle, including the fire extinguishing equipment, and any special attachments or auxiliary equipment. The operator's manual shall include at least the following:

a. Location and function of all controls and instruments shall be illustrated and fully described.

b. Safety information that is consistent with the safety standards established by the Occupational Safety and Health Administration (OSHA).

c. Checks and adjustments in preparation for placing the vehicle for service upon receipt from the contractor.

d. Preparation for shipment or storage.

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e. Warranty information and period of the warranty for the complete vehicle and for any component warranty that exceeds the warranty of the complete vehicle. Addresses and telephone numbers shall be provided for all warranty providers.

f. General description of and step-by-step instructions for the operation of the vehicle and its fire extinguishing system(s) and auxiliary equipment.

g. Description of the post-operational procedures (draining, flushing, et cetera).

h. Checklists for the daily maintenance inspection and mission readiness checks that the operator is expected to perform, including basic troubleshooting procedures.

i. Procedures for towing a disabled vehicle.

j. Procedures for changing a tire.

k. Schedules for required preventative maintenance and required periodic maintenance.

l. A line art drawing of the vehicle, including front, rear, left, and right side views, showing basic dimensions and weights (total vehicle and individual axle weight for the unloaded and fully loaded vehicle).

m. Procedures and maximum speed limits for towing a trailer, including hazards, risks, and special requirements.

2.1.7.1.2 Service manual. The service manual shall identify any special tools and test equipment required and shall cover troubleshooting and maintenance as well as minor and major repair procedures. The text shall contain performance specifications, tolerances, and fluid capacities; current, voltage, and resistance data; test procedures; and such illustrations and exploded views as may be required to permit proper maintenance by qualified mechanics. The manual shall contain an alphabetical subject index as well as a table of contents. The service manual shall contain at least the following, where applicable:

a. Fire fighting system schematic(s).

b. Hydraulic schematic.

c. Pneumatic schematic.

d. Electrical schematic.

e. Winterization schematic.

f. Fuel schematic.

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g. Schedules for required preventative maintenance and required periodic maintenance.

h. Location, procedure, and interval for parts of the truck and equipment that require lubrication.

2.1.7.1.3 Parts manual. The parts manual shall include illustrations and exploded views, as needed, to properly identify all parts, assemblies, subassemblies, and special equipment. All components of assemblies shown in illustrations or exploded views shall be identified by reference numbers that correspond to the reference numbers in the parts lists. All purchased parts shall be cross-referenced with the original manufacturer's name and part number. The parts identification manual shall provide the description and quantity of each item used per vehicle. The size, thread dimensions, torque specifications, and special characteristics shall be provided for all nonstandard nuts, bolts, screws, washers, grease fittings, and similar items. The manual shall contain a numerical index. The parts manual shall contain a list of all of the component vendor names, addresses, and telephone numbers referenced in the parts list.

2.1.8 Painting, plating, and corrosion control.

2.1.8.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be in contact with each other. Metal plating or metal spraying of dissimilar base metals to provide electromotively compatible abutting surfaces is acceptable. The use of dissimilar metals separated by suitable insulating material is permitted, except in systems where bridging of insulation materials by an electrically conductive fluid can occur.

2.1.8.2 Finish. Exterior surfaces shall be prepared, primed, and painted with polyurethane paint in accordance with all of the paint manufacturer's instructions and recommendations. Unless otherwise specified (see 6.2), the exterior finish color shall be Candy Apple Red, Sikkens Color Number FLNA3021, DuPont Color Number 97902U or 4737U, PPG Color Number 71528 (the PPG name for this color is Cardinal Red), or equal. When specified, the exterior finish color shall be Desert Sand, Color Number 30313 of FED-STD-595. The interior of all compartments shall be painted with an impact resistant, textured coating that resists stains, scuffs, chips, and scratches.

a. For vehicles painted Candy Apple Red, the cab upper body (from the bottom of the windshield) and roof shall be painted White, Color Number 17875 of FED-STD-595. All bright metal and anodized parts, such as mirrors, horns, light bezels, treadplate, and roll-up compartment doors, shall not be painted. Compartment interiors shall be gray.

b. For vehicles painted Desert Sand, all exterior surfaces, excluding all normally bright metal and anodized parts and any interior surfaces visible with any compartment door open (but not the interior of the cab), shall be painted the color of the body. This includes compartment shelves and mounting hardware, but does not include items mounted in the compartments. Non-metallic materials may be black or gray.

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2.1.8.3 Reflective stripes. Horizontal, reflective stripes in accordance with 15.9.3 of NFPA 1901 shall be applied around the vehicle in an approximate plane with the headlights. Offsets in the reflective stripes shall be made to maximize the length of reflective surface. Bright metal trim or anodized parts may interrupt the reflective stripes. The pattern shall be 10 inches wide with three reflective stripes (one inch reflective, one inch body color, six inches reflective, one inch body color, and one inch reflective). The reflective stripes shall be white for vehicles painted Candy Apple Red and Desert Sand.

2.1.8.4 Chevron Striping. In accordance with 15.9.3.2 of NFPA 1901 at least 50 percent of the rear-facing vertical surfaces, visible from the rear of the apparatus, excluding any pump panel areas not covered by a door, shall be equipped with retroreflective striping in a chevron pattern sloping downward and away from the centerline of the vehicle at an angle of 45 degrees.

2.1.8.4.1 Each stripe in the chevron shall be a single color alternating between red and fluorescent yellow in accordance with 15.9.3.2.1.

2.1.8.4.2 Each stripe shall be 6-inches (150 mm) in width in accordance with 15.9.3.2.2 of NFPA 1901 and the example provided below.



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2.1.8.5 Lettering. Vehicles painted Candy Apple Red shall have the letters "UNITED STATES" and "AIR FORCE" applied in synthetic or encapsulated gold leaf, with outline and black shadow, on the front door on both sides in long radius elliptical arches above and below the lettering center line. The size of the lettering shall be a minimum of 2½-inches to a maximum of 6-inches.

2.1.9 Identification plate. A permanently marked identification plate shall be securely mounted at the driver's compartment. The identification plate shall contain the following information:

a. NOMENCLATURE
b. MANUFACTURER'S MAKE AND MODEL
c. MANUFACTURER'S SERIAL NUMBER
d. REGISTRATION NUMBER
e. NATIONAL STOCK NUMBER (NSN)
f. VEHICLE CURB WEIGHT: kg (pounds)
g. PAYLOAD, MAXIMUM: kg (pounds)
h. GROSS VEHICLE WEIGHT (GVW): kg (pounds)
i. FUEL CAPACITY AND TYPE: gal (gallons) / L (liters)
j. DATE OF DELIVERY (month and year)
k. WARRANTY (months and km (miles))
l. CONTRACT NUMBER
m. PAINT COLOR AND NUMBER
n. VEHICLE LENGTH, WIDTH AND HEIGHT (in / cm)

2.2 General requirements.

2.2.1 Hose storage.

2.2.1.1 Hose storage area. The tanker body shall have a hose bed, with one or more removable covers, in accordance with 7.5 of NFPA 1901. Hose bed covers shall be red in color for vehicles painted Candy Apple Red. For vehicles painted Desert Sand, hose bed covers shall be tan in color. Covers will be secured by snap fasteners.

2.2.1.2 Cross lay hose beds. The tanker shall have two cross lay hose beds, between the back of the cab and the front of the water tank, each capable of carrying a 200 foot preconnected 1¾-inch double jacketed fire hose. The hose beds shall be equipped with one or more removable covers. Sufficient clearance shall be provided between the hose and cover to allow a preconnected hose nozzle to be pulled through from either side of the tanker. Cross lay hose bed covers shall be red in color for vehicles painted Candy Apple Red. For vehicles painted Desert Sand, cross lay hose bed covers shall be tan in color. Covers will be secured by snap fasteners.

2.2.2 Suction hose.

2.2.2.1 Hard suction hose. Two 10 foot long, light weight 6-inch hard suction hoses, with 6-inch National Hose thread long handle female couplers on one end and 6-inch National Hose thread long handle male couplers on the other, shall be mounted on the tanker. A suction strainer shall

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also be mounted on the tanker. The hose and suction strainer mounting system design shall be subject to approval by the procuring activity.

2.2.2.2 Soft suction hose. The tanker shall be equipped with a minimum 15 foot long, 6-inch soft suction hose, with a 4½-inch National Hose thread long handle female coupler on one end and a 6-inch National Hose thread long handle female coupler on the opposite end. The soft suction hose and couplers shall be stored in a compartment.

2.2.3 Minor equipment.

2.2.3.1 Fire hose and nozzles. The tanker shall be designed to accommodate purchaser provided nozzles and hose listed in 7.5 – 7.7 of NFPA 1901, stored in compartments. The fire hose and nozzles shall not be provided with the tanker.

2.2.4 Equipment. The tanker shall have compartment space for and also the full complement of equipment, except for the self-contained breathing apparatus (SCBA) and the spare SCBA cylinders. Two individual compartments, or a two bottle rack in a compartment, shall be included for two purchaser provided one hour capacity SCBA air bottles**.**

2.2.5 Chocks. Two folding wheel chocks, as required by 7.7 of NFPA 1901 shall be provided and mounted in readily accessible locations.

2.2.6 Portable water storage pond. The tanker shall be equipped with a 5,000 gallon minimum capacity portable water storage pond/emergency water supply storage tank. The pond wall height shall not exceed the height of the bottom of the water tank dump valve, while the tanker and pond are on a common level surface. The pond shall be stored in a compartment that provides at least 4-inches of clearance on all sides and the top of the pond. The compartment used for storage of the portable water storage pond shall be appropriately marked to indicate that multiple personnel are required for removal and reloading due to the weight of the pond.

2.3 Chassis and vehicle components.

2.3.1 Capacity, drive train and chassis. The tanker shall have a net payload capacity of 4,000 gallons of water plus 1,000 pounds for crew and equipment. The chassis will include a heavy-duty driveline assembly (including heavy-duty axles and suspension) rated for “on-highway and off-road” usage consistent with at least the net payload specified above. The suspension system shall include rear axle torque rod/axle configuration which allows for vehicle flexibility to traverse uneven off-road terrain.

2.3.2 Engine. The tanker shall have a turbocharged diesel engine that is certified to comply with the Environmental Protection Agency (EPA) on-highway emission requirements at the time of manufacture. When specified (see 6.2), the engine shall run on diesel rather than Ultra Low Sulfur Diesel.

2.3.2.1 Fuel filters. Primary and secondary fuel filters and a fuel/water separator shall be provided. The fuel/water separator shall include a water coalescer and a drain valve that is

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readily accessible by an operator or a mechanic and shall be in accordance with SAE J1839. A combination fuel filter and fuel/water separator may be provided. Fuel filter elements shall be easily replaceable by a mechanic without loss of engine prime.

2.3.3 Exhaust system. The exhaust system outlet(s) shall be directed away from personnel accessing equipment compartments or the tank manhole access ladder.

2.3.4 Brake system. The tanker shall be equipped with an all-wheel antilock brake system; the brakes shall be fully air-actuated. Brakes shall be in accordance with Code of Federal Regulations (CFR) 49 CFR 393.40 through 393.42(b) and 393.43 through 393.52. The braking system complete with all necessary components shall include:

a. Air compressor having a capacity of not less than 12 standard cubic feet per minute (scfm).

b. Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between the compressor and the reservoir tank. Tank drains shall be appropriately labeled.

c. At least one automatic moisture ejector shall be provided in the system.

d. Automatic slack adjusters on cam brakes or internal self-adjusting brakes on wedge and disc brakes on all axles.

e. Spring set parking brakes.

All components of the braking system shall be installed in such a manner as to provide adequate road clearance when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake system components. No part of the braking system shall extend below the bottom of wheel rims, to ensure, in case of a flat tire, that the weight of the vehicle will be supported by the rim and the flat tire and not be imposed on any component of the braking system. Slack adjusters and air chambers shall be located above the bottom edge of the axle carrier.

2.3.4.1 Air dryer. A replaceable cartridge desiccant air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95% of the moisture in the air being dried. The dryer shall have a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

2.3.4.2 Compressed air shoreline. A checked, auto-eject compressed air shoreline connection shall be provided to maintain brake system pressure while the vehicle is not running. The shoreline shall be flush mounted (not to extend outside the body line), located on the exterior of the vehicle, either on the left side rear corner of the cab or at the pump panel. A minimum 50 foot long air supply hose equipped with an appropriate mating shoreline connector and an air fitting shall be provided with the vehicle.

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2.3.4.3 Auxiliary braking system. The service brakes shall be augmented by one of the following auxiliary braking systems:

a. A system which opens all or some of the engine exhaust valves near the end of the compression stroke, thereby converting vehicle motion to a pumping loss. The engine brake shall be approved by the engine manufacturer.

b. A controlled gate valve in the exhaust manifold, which produces engine back pressure. The exhaust brake shall be approved by the engine manufacturer.

c. A hydrodynamic retarder integral with the transmission.

A dash mounted switch shall be provided to activate, modulate, or cut out the brake augmentation. The switch shall be marked to indicate its position. When active, the system shall be fully controlled by means of the conventional driving controls to apply retardation during vehicle deceleration, and cut out in the other operating modes.

2.3.5 Tires and wheels. The tanker shall be equipped with tubeless steel belted radial tires with highway type tread rated for specified highway speeds (see 2.1.3.3) as well as off-road conditions (see 2.3.1 above), mounted on steel disc wheel assemblies. If all tire and wheel assemblies are identical, one spare tire and wheel assembly shall be provided. If two different tire and wheel assemblies are provided, two spare tire and wheel assemblies shall be provided, one of each configuration. Spare tire tread patterns, size, and type will be the same as those mounted on the vehicle. Spare tires and wheels are not required to be mounted on the tanker. Tire valve stem caps shall be made of plastic.

2.3.6 Tire chains. When specified (see 6.2), the tanker shall be equipped with driver activated automatic type tire chains for the rear axles.

2.3.7 Fenders. Rear fenders and fender liners having tire chain clearance shall be provided.

2.3.8 Steering. The tanker shall be equipped with power steering.

2.3.9 Transmission. A fully automatic transmission with a hydraulic torque converter and at least four forward speeds shall be provided. The normal driving range selector position shall provide at least four gear ratios without movement of the selector. The transmission shall include an oil cooling system capable of limiting the transmission temperature to the maximum recommended by the transmission manufacturer. The transmission shall have a direct drive lockup for pumping.

2.3.10 Traction control. The tanker shall be equipped with limited slip or automatic locking differentials, including a driver-controlled capability to lock the two tandem axles together for increased rear traction as conditions warrant.

2.3.11 Fuel tank. Fuel tank(s) having a minimum usable capacity in accordance with 12.3.4 of NFPA 1901 or 50 gallons, whichever is greater, shall be provided. The location of the fuel

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tank(s) shall protect the tank(s) from mechanical damage during normal use of the tanker. Signage will be provided at the fill point to indicate “Diesel Fuel Only” or “Ultra Low Sulfur Fuel Only”.

2.3.12 License plate bracket. A lighted license plate bracket shall be provided at the left rear of the tanker.

2.3.13 Trailer towing package. The tanker shall be equipped with a Class III tow hitch with a square sleeve type receiver used to tow trailers and other ancillary fire fighting equipment. The tow hitch and receiver shall be rated at the maximum inherent towing capability of the tanker such that safety and reliability of the tanker are not degraded. The maximum towing capacity and tongue weight capacity of the tanker shall be clearly and visibly posted both near the tow hitch receiver and inside the cab of the tanker. The manufacturer shall describe any hazards, risks, maximum suggested speed limits or restrictions, and special use requirements during the solicitation process. This information shall also be listed near the tow hitch receiver, inside of the cab of the tanker, as well as in the operator's manual. A brake controller shall be provided for use with a trailer which has an electric braking system. Gladhands in accordance with SAE J318 shall be provided for use with a trailer which has an air braking system.

2.3.13.1 Safety chain attachment points. The tanker shall be equipped with two safety chain attachment points, one located on each side of the receiver sleeve, symmetrical about the receiver sleeve. The vertical centerline of the of the safety chain attachment points shall be located at a parallel distance of not greater than 6 inches to the left and right of the tanker centerline. Each of the attachment points shall be capable of withstanding a maximum shock load of 10% of the maximum vehicle towing capacity and shall have the ability for standard safety chain hooks to be attached. The inside radius of the safety chain hook is between 1 inch and 3 inches.

2.3.13.2 Electrical harness receptacle. The tanker shall have a weatherproof 7-way round (pin) electrical receptacle installed within 12 inches of the receiver sleeve. The electrical receptacle shall have provisions to provide electrical power to a trailer with electric brakes. All terminals of the receptacle shall be wired in accordance with SAE J560b. A weatherproof 7-way flat (blade) electrical receptacle shall be included as loose equipment. A heavy-duty flasher designed for trailer applications shall be provided to support the additional electrical load of the trailer's lighting system.

2.4 Low-voltage electrical systems. The tanker shall have a 12-volt electrical and starting system.

2.4.1 Alternator. A single alternator charging system in accordance with 13.3 of NFPA 1901 shall be provided. The minimum continuous electrical load shall include operation of the air conditioning system.

2.4.2 Batteries. Batteries shall be of the maintenance-free type; addition of water shall not be required during normal service life. The battery cover and vent system shall be designed to prevent electrolyte loss during service and to keep the top of the battery free from electrolyte.

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2.4.2.1 Battery compartment. The batteries shall be mounted in an acid-resistant tray provided with replaceable pads and shall be enclosed in a weatherproof box or compartment.

2.4.3 Battery charger or conditioner. The tanker shall have a DC taper type battery charger or an automatic battery conditioner, providing a minimum 12 amp output. The charger/conditioner shall be permanently mounted on the tanker in a properly ventilated, accessible location. The charger/conditioner shall be powered from the electrical shoreline receptacle (see 2.9.1). A charging indicator shall be installed next to the receptacle. When a battery conditioner is provided, the conditioner shall monitor the battery state of charge and, as necessary, automatically charge or maintain the batteries without gassing, depleting fluid level, overheating, or overcharging.

2.4.4 Emergency warning lights. All emergency warning lights shall use strobe type or light emitting diode (LED) elements; LEDs should be used where available. The emergency warning light system, related components, and devices shall be in accordance with 13.8 of NFPA 1901.

2.4.4.1 Light bar. A light bar with both forward and side facing strobe or LED elements shall be mounted on or above the cab roof. Forward facing lenses or LED elements shall be red or red and white. If rearward facing lenses or LED elements are provided (note that NFPA 1901 requires that warning lights be constructed or arranged so as to avoid projection of the light into any driving or crew compartment), they shall be red or red and amber. The light bar shall be separately switched from the warning light panel. When specified (see 6.2), trucks shall be equipped with a steady red front warning light as required by California state law.

2.4.4.2 Emergency warning light color. When specified (see 6.2), the rearward, red emergency warning lights shall be replaced with amber. When specified (see 6.2), all red emergency warning lights shall be replaced with blue. When blue lights are specified, the optical power requirements of NFPA 1901 are waived.

2.4.4.3 Headlight flashing system. A high beam, alternating/flashing, headlight system shall be provided. The headlight flasher shall be separately switched from the warning light panel.

2.4.5 Audible warning devices.

2.4.5.1 Siren. The tanker shall be equipped with an electronic siren system. The amplifier unit shall include volume control and selection of "Radio," "PA," "Manual," "Yelp," "Wail," and "Hi-Lo" (European) modes, and a magnetic noise canceling microphone. The amplifier, microphone, and controls shall be within reach of the driver and right front crew member. Siren activating foot switches shall be located in front of the driver and the right front crew member. The siren speaker shall be rated at 100 watts minimum and shall be located in a guarded position as low and as far forward on the vehicle as practical.

2.4.5.2 Horn. Dual forward facing air horns shall be installed in protected locations near the front of the truck. Air horn activating foot switches shall be located in front of the driver and the right front crew member.

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2.4.6 Work lighting.

2.4.6.1 Cab interior lights. Cab interior light levels shall be a minimum of 2 foot-candles.

2.4.6.2 Compartment lights. White lighting sufficient to provide an average minimum illumination of 1 foot-candle shall be provided in each compartment greater than 4 cubic feet and having an opening greater than 144 square inches. Where a shelf is provided, this illumination shall be provided both above and below the shelf. Lights shall automatically illuminate only when the respective door is opened and the vehicle electrical master switch is in the "on" position. Light switches shall be of the magnetic (non-mechanical) type.

2.4.6.3 Ladder, step, walkway, and area lights. Non-glare white or amber lighting shall be provided at ladders and access steps where personnel work or climb during night operations. These area lights shall be controlled with switches on the cab instrument panel and near the light sources.

2.4.6.4 Spotlights. Two halogen spotlights rated at 160,000 candlepower (CP) minimum each shall be mounted on the cab roof and controlled from inside the cab.

2.4.7 Scene lights. A total of 8 12-volt High Intensity Discharge (HID) high-mounted floodlights shall be provided to illuminate the work areas around the tanker: two on each side, two at the rear, and two in the front. Individual switches shall be located in the work areas and on the instrument panel. A switch shall be provided to control the lights on the left side of the tanker, a second switch shall be provided to control the lights on the right side of the tanker, and a third switch shall be provided to control the lights at the rear of the tanker. A separate switch for the front lights shall be located on the instrument panel only. Three way switches shall be used.

2.4.8 Radio circuit. The vehicle shall have two separate 30 amp circuits, with breakers and at least 6-foot of wire routed to a space provided adjacent to the driver and turret operator for purchaser provided radios and other electrical equipment. The wiring shall be tagged indicating its purpose.

2.4.9 Auxiliary power receptacles. The tanker shall have 2 12-volt auxiliary power receptacles mounted adjacent to the driver and crew member positions, preferably in the instrument panel or in a center radio mounting console.

2.5 Driving and crew areas.

2.5.1 Cab. The tanker shall have a two person, two door, non-tilting cab. A tilt steering column shall be provided.

2.5.1.1 Seats. The driver's seat shall be of an adjustable air suspension design. The crew member seat shall be of a non-suspension design with a backrest and brackets designed to store a one-hour capacity SCBA. Each seat shall be provided with a Type 2 seat belt assembly (i.e., 3-point restraint) in accordance with Federal Motor Vehicle Safety Standard (FMVSS) 209.

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2.5.1.2 Cab interior sound level. The maximum sound level at any seat location shall not exceed 90 dBA without any warning devices in operation, as measured in accordance with 49 CFR 393.94(c), "Vehicular interior noise levels test procedure," except that the test shall be performed with the vehicle traveling at a steady speed of 45 mph on a level, hard, smooth surface road.

2.5.1.3 Windshield and windows. The windshield and windows shall be of tinted safety glass in accordance with industry standards.

2.5.1.4 Mirrors. Outside rearview mirrors shall be installed on each side of the cab, mounted on fold-back brackets. The mirrors shall also have electrically heated heads. Mirror heating controls shall be located on the instrument panel within reach of the seated driver.

2.5.1.5 Climate control system. The offeror/contractor's standard heater/defroster and air conditioning system shall be provided.

2.5.1.6 Instruments and controls. Gauges shall be provided for oil pressure, coolant temperature, and automatic transmission temperature. All switches and/or controls that activate systems or system components shall be labeled as to their function. In addition to the instruments and controls required by 14.3.6 of NFPA 1901, the following shall be provided within convenient reach of the seated driver:

a. Master warning light control switch,

b. Work light switch(es), and

c. Compartment "Door Open" warning light and intermittent alarm that sounds when a compartment door is open and the parking brakes are released or the transmission is in any position other than neutral.

d. Lateral G-Force Indicator

2.6 Body, compartments, and equipment mounting.

2.6.1 Body. The tanker shall have an all aluminum or all stainless steel body. Aluminum thickness shall be at least 0.125 inches; stainless steel thickness shall be at least 0.090 inches. The tanker body may or may not enclose the water tank.

2.6.2 Compartments. The tanker body shall include sufficient compartment space to hold all of the equipment listed in 2.2.3.

2.6.2.1 Compartment doors. Storage compartments shall have clear anodized aluminum, counterbalanced, non-locking, roll-up doors. Door latch handles shall be full-width bar type. Door straps shall be provided to assist in closing the compartment doors when the rolled up door height exceeds 6 feet above the ground.

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2.6.2.2 Scuffplates. Replaceable scuffplates shall be provided at each compartment threshold to prevent body damage from sliding equipment in and out of the compartments. The scuffplates shall be securely attached to the compartment threshold but shall be easily replaceable in the event of damage.

2.6.2.3 Drip rails. Drip rails shall be provided over each compartment door. If the drip rails are not integral with the body, they shall be of anodized extruded aluminum and shall have a bright finish for vehicles painted Candy Apple Red.

2.6.2.4 Shelves. An adjustable and removable compartment shelf shall be provided for every 18 inches of each vertical storage compartment door opening. Shelving adjustments shall require no more than common hand tools, and shall not require disassembly of fasteners. Shelves shall support a minimum of 200 pounds without permanent deformation. Each shelf shall be accessible to crew members standing on the ground or using a pull out and tip-down configuration for shelving over 54 inches from the ground. Each shelf shall have ¼ inch diameter drain holes located in each corner of the shelves so as to allow for drainage of any water from the stowed equipment.

2.6.2.5 Drainage mats. Each compartment floor and shelf shall be covered with a removable black mat designed to allow for drainage of any water from the stowed equipment.

2.6.3 Ladder, handrails, and walkways. The tanker shall have a ladder and walkway to access the manhole(s) and servicing points along the top of the water tank. Folding handrails or tie-on points shall be provided to restrain personnel from falling to the ground. All ladders, stepping, standing, and walking surfaces shall be in accordance with 15.7 of NFPA 1901. Handrails shall be provided in accordance with 15.8 of NFPA 1901.

2.7 Fire pump and associated equipment.

2.7.1 Fire pump. The fire pump shall be a single stage centrifugal design, with a thick walled cast iron pump, a bronze impeller and a stainless steel shaft, capable of providing 1,250 gpm of water at 150 psi. The cast iron pump body must not contribute in any way to the premature failure of the fire pump. The pump shall be driven from the chassis propulsion engine through a split shaft power takeoff with ratings to handle the full torque and power applied.

2.7.2 Piping and associated components. All metallic surfaces of the piping and associated components that come into contact with the water shall be of passivated stainless steel or bronze and shall be capable of storing brackish/saltwater. The discharge piping shall flow water at a minimum of 1,250 gpm.

2.7.3 Intake connections. The tanker shall have two 6-inch intake connections, one on each side near the front of the tank. The tanker shall have two gated 2½-inch intake connections, one on each side of the tanker, adjacent to the 6-inch intake connections and fitted with 30° or 45° turn-down fittings. Each intake connection shall be gated and shall have National Hose threads.

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2.7.4 Discharge connections. The tanker shall have two 4-inch minimum discharge connections, one on the right side of the tanker near the front of the tank and one at the rear of the tanker on the driver's side. A 4-inch discharge shall not be provided at the pump operators control panel on the left side of the truck. The tanker shall have two 2½-inch discharge connections, one on each side of the tanker near the front of the tank. Each 2½-inch discharge connection shall be equipped with no less than 3-inch full flow piping and valve with a 2½-inch adapter. Each side discharge connection shall be fitted with a 30° or 45° turn-down fitting. The tanker shall be equipped with two 2-inch preconnected swivel discharge outlets to accommodate the cross lay hoses. All discharge connections shall have National Hose threads.

2.7.5 Pump operator control panel. A pump operator control panel in accordance with 16.9 of NFPA 1901 shall be mounted on the left center side of the tanker. Gauges shall be provided. The panel shall have master intake and discharge pressure gauges, not less than 6-inch diameter, and individual discharge outlet pressure gauges, not less than 2-inch diameter. Each discharge outlet shall be controlled from the pump operator panel through a manual valve having locking features. The water tank level indicator shall be of the five light type; that is, the indicator shall have lights to show the water tank level at quarter-tank-capacity intervals and have a flashing "refill" or "tank empty" light. Discharge and intake controls shall be color coded in accordance with 16.9.1 of NFPA 1901. A green "battery on" pilot light shall be mounted on the pump operator control panel.

2.8 Water tank. The tanker shall have a water tank with a manufacturers certified capacity of at least 4,000 gallons.

2.8.1 Water tank material. The water tank shall be constructed of polypropylene.

2.8.2 Water tank baffling. The water tank shall be designed for dynamic baffling in accordance with 18.2 of NFPA 1901.

2.8.3 Tank to pump intake line. All metallic surfaces of the tank to pump intake line and associated components that come into contact with the water shall be of passivated stainless steel or bronze and shall be capable of storing brackish/saltwater. The tank to pump intake line shall be capable of delivering water at a flow rate in accordance with 18.3 of NFPA 1901 for the certified capacity of the water tank provided. The tank to pump valve(s) shall be air or electrically actuated with a manual override.

2.8.4 Manhole. The water tank shall have a manhole of not less than 20 inches inside diameter for personnel access to the interior of the tank for cleaning and inspection.

2.8.5 Tank fill connection. A 6-inch gated fill connection shall be located at the right rear of the water tank.

2.8.6 Dump valve. Pneumatic dump valves shall be installed at the rear face and rear left and rear right sides of the water tank for filling the portable pond provided with the tanker. The dump valve shall empty 90% of the water tank volume within 3 minutes with the tanker on a level surface. A rigid folding chute or flexible tube extension shall be provided to drain the water in

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the tank into the portable pond. The chute or extension shall be stored in a readily accessible location and shall be easily attached to the dump outlet. Controls for the dump valve shall be installed both near the outlet and in the cab in a location readily accessible to the seated driver.

2.9 Line voltage electrical system.

2.9.1 Electrical shoreline connection. The battery charger/conditioner shall be powered from a covered, three wire, straight blade, polarized, insulated, labeled, recessed, 120 volt, NEMA 5-15P male plug inlet, in a weatherproof AC auto-eject receptacle. The connection shall be located on the exterior of the vehicle, either on the left side rear corner of the cab or within 6 inches of the left side front corner of the body. A 50 foot long, three wire, 15 amp rated, 120 volt, AC power cable, with straight blade (non twist-lock), NEMA 5-15R and 5-15P style, in accordance with NEMA WD-6, connectors, shall be provided. When equipped with a winterization system, two identical receptacles and cables may be provided; the receptacles shall be clearly marked. The winterization system shall have a labeled activation switch with a pilot light. When specified (see 6.2), the receptacle(s) and power cable(s) shall be for 220 volts in lieu of 110 volts.

2.10 Workmanship. The tanker, including all parts and accessories, shall be fabricated in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, burrs, defects, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of welding, brazing, soldering, riveting, and painting; alignment of parts; tightness of fasteners; et cetera. The tanker shall be thoroughly cleaned of all foreign matter.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

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4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First production inspection. The first production tanker shall be subjected to the examinations and tests described in 4.6.3.1 through 4.6.3.6. The contractor shall provide or arrange for all test equipment and facilities.

4.3 Conformance inspection. Each production vehicle shall be subjected to the examinations and tests described in 4.6.3.1, 4.6.3.2.1 through 4.6.3.2.4, 4.6.3.3, 4.6.3.5, and 4.6.3.6.

4.4 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

4.5 Commercial item requirement. The vehicle furnished shall comply with the "commercial item" definition of FAR 2.101 as of the date of award. The Government reserves the right to require the offeror/contractor to prove that their product complies with the referenced commerciality requirements and each salient characteristic of this CID. The offeror/contractor shall provide an itemized technical proposal that describes how the proposed model complies with each salient characteristic of this CID; **a paragraph by paragraph response to the salient characteristics section of this CID shall be provided**. The proposal shall be provided with the pricing submission. Failure to provide this information may deem a vendor as non-responsive and their proposal may be rejected. The offeror/contractor shall provide two copies of their commercial descriptive catalogs with their offer as supporting reference to the itemized technical proposal. The offeror/contractor shall identify all modifications made to their commercial model in order to comply with the requirements herein.

4.6 Inspection requirements.

4.6.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

4.6.2 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

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b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

h. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

i. Overheating of the engine, transmission, or any other vehicle component.

j. Evidence of corrosion.

k. Failure of the firefighting system.

4.6.3 Detailed inspection requirements.

4.6.3.1 Examination of product. Each tanker shall be examined to verify compliance with the requirements herein. A Government generated checklist that identifies each relevant requirement and the inspection results shall be used. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each tanker function shall be verified. Each production tanker shall be inspected to a reduced version of the checklist.

4.6.3.2 Road tests. The following tests shall be performed in accordance with 4.17 of NFPA 1901 and shall include traversing 5 miles of gravel roads.

4.6.3.2.1 Acceleration test. Each tanker shall be tested to demonstrate compliance with 2.1.3.2. For the first production truck, a time-distance recorder shall be used to record data for this test.

4.6.3.2.2 Maximum speed test. Each tanker shall be tested to demonstrate compliance with 2.1.3.3. For the first production unit, a time-distance recorder shall be used to record data for this test.

4.6.3.2.3 Auxiliary braking system test. Each tanker shall be tested in accordance with 4.17.7 of NFPA 1901.

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4.6.3.2.4 Service brake system test. Each tanker shall be tested in accordance with 4.17.8 of NFPA 1901. For the first production vehicle, a time-distance recorder shall be used to record data for this test.

4.6.3.2.5 Turning diameter test. The fully loaded first production tanker shall be tested as follows to demonstrate compliance with 2.1.5.

a. Facilities. This test shall be performed on a dry, level, paved area that is free from loose material and is larger in all directions than three times the length of the vehicle being tested.

b. Equipment Required.

(1) A device suitable for measuring three times the length of the vehicle being tested with an accuracy of at least ±1-inch.

(2) Markers or marking device suitable for marking the pavement.

(3) A plumb bob suitable for locating a point on the pavement directly below a fixed point on the vehicle.

(4) A test report notebook or similar record forms to be used as a test report work sheet and incorporated into the documentation package.

c. Test Conditions.

(1) The vehicle shall be fully loaded. Ballast shall be used as needed to account for the crew and equipment allowance. The ballast shall be arranged so as to distribute the weight in a manner that closely simulates the items being represented. The ballast shall not be shaped or distributed in a manner that creates a favorable, artificial center of gravity.

(2) The vehicle steering system shall be fully operational, and the steering linkage stops shall be adjusted to the manufacturer's specified production tolerance limits.

d. Test Procedure.

(1) The vehicle shall be driven slowly in a full cramp circle (left or right) to establish a steady state in the steering linkage.

(2) Continue driving the slow full cramp circle.

(3) At approximately three equidistant points (identified as A, B, and C) around the circle, gently stop the vehicle using the service brakes.

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(4) At each stop, place a plumb bob against the outermost point of the vehicle and mark the spot on the ground directly below where the plumb bob comes to rest.

(5) Measure and record the straight line distances between each pair of points, i.e., Lengths AB, BC, and CA.

(6) Calculate the wall-to-wall turning diameter (D) as follows:

Length S = [AB + BC + CA] / 2

D = [AB x BC x CA] / {2 x sqrt[S x (S-AB) x (S-BC) x (S-CA)]}

(7) Repeat steps (1) through (6) with the vehicle moving in the opposite direction.

4.6.3.2.6 Mobility test. The fully loaded first production tanker shall be driven over 10 miles of paved roads and five miles of gravel roads. All loads shall be removed and all structure and surfaces shall be visibly inspected for failure or permanent deformation.

4.6.3.2.7 Weights and Measurements. The vehicle shall be weighed in both the empty and full configuration. Length, width, and height measurements shall be validated in accordance with 2.1.4.

4.6.3.2.8 Ancillary Equipment: Each tanker shall be inventoried to ensure equipment listed in 7.7 of NFPA 1901 is provided. Equipment shall not be mounted to the vehicle. It will be packaged and shipped as lose equipment.

4.6.3.3 Low voltage electrical system performance tests. Each tanker shall be tested in accordance with 13.14 of NFPA 1901.

4.6.3.4 Cab interior sound level test. The cab interior sound levels of the first production tanker shall be measured in accordance with 49 CFR 393.94(c), "Vehicular interior noise levels test procedure," except that the test shall be performed with the vehicle traveling at a steady speed of 45 mph on a level, hard, smooth surface road.

4.6.3.5 Test of fire pump and associated equipment. Each tanker shall be tested in accordance with 16.13 of NFPA 1901. The costs associated with the independent certification shall be included in the price of the vehicle.

4.6.3.6 Line voltage electrical system testing. Each tanker shall be tested in accordance with 22.15 of NFPA 1901.

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5. PACKAGING.

5.1 Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Department of Defense and Federal documents may be obtained at http://assist.daps.dla.mil or from the Document Automation and Production Service, Bldg 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia PA 19111-5094.

6.1.2 The Code of Federal Regulations (CFR) may be obtained at http://www.gpoaccess.gov/cfr/ or from the Superintendent of Documents, U.S. Government Printing Office, Washington DC 20402.

6.1.3 SAE documents may be obtained http://www.sae.org/servlets/index or from SAE, Inc., 400 Commonwealth Drive, Warrendale PA 15096

6.1.4 NFPA documents may be obtained at http://www.nfpa.org/index.asp or from NFPA, Batterymarch Park, Quincy MA 02269-9101.

6.2 Ordering data. The contract or order should specify the following:

a. If tanker will be operated at altitudes greater than 2,000 feet above sea level (see 2.1.1.1).

b. Finish color required (Desert Sand in place of Candy Apple Red) (see 2.1.8.2).

c. If diesel fuel is required in lieu of ultra low sulfur diesel fuel.(see 2.3.2)

d. If driver activated automatic type tire chains are required (see 2.3.6).

e. If a steady red front warning light is required (see 2.4.4.1). (NOTE: Applies to deliveries in California only)

f. Emergency warning light color required (amber or blue in place of red) (see 2.4.4.2) (NOTE: Applies to USAFE ONLY).

g. Electrical shoreline voltage, 220 volts instead of 110 volts (see 2.9.1) (NOTE: Applies to some USAFE and South West Asian (CENTAF) locations only).

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6.3 Key Words.

Brush
Gravel roads
Paved roads
Unimproved surfaces

Custodian:
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Reviewer:
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Agent:
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